



## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | AHT30_lib          |
| Board Name      | custom             |
| Generated with: | STM32CubeMX 6.15.0 |
| Date            | 06/09/2026         |

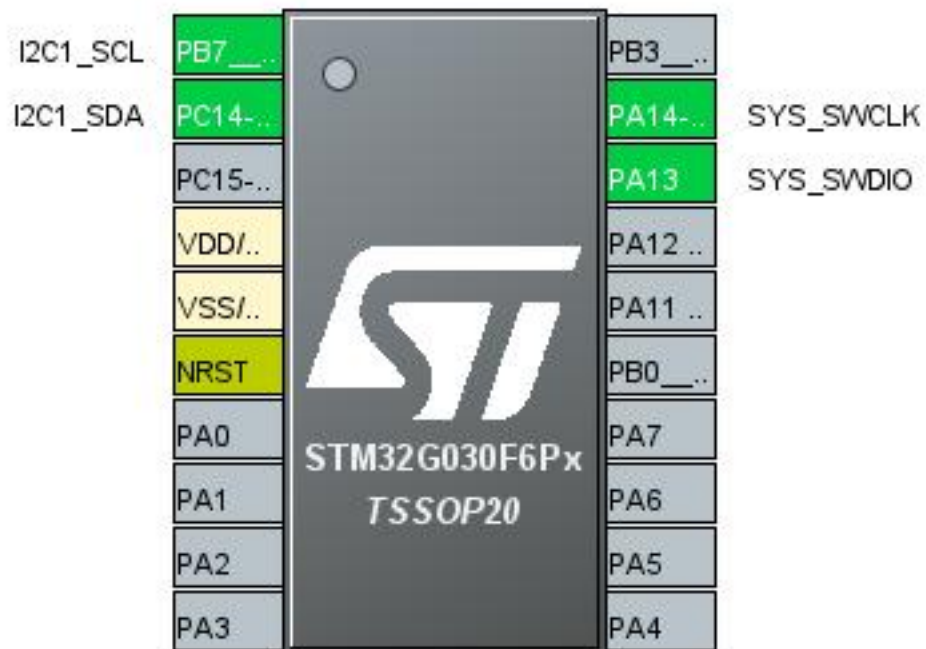
### 1.2. MCU

|                |                      |
|----------------|----------------------|
| MCU Series     | STM32G0              |
| MCU Line       | STM32G0x0 Value line |
| MCU name       | STM32G030F6Px        |
| MCU Package    | TSSOP20              |
| MCU Pin number | 29                   |

### 1.3. Core(s) information

|         |                |
|---------|----------------|
| Core(s) | ARM Cortex-M0+ |
|---------|----------------|

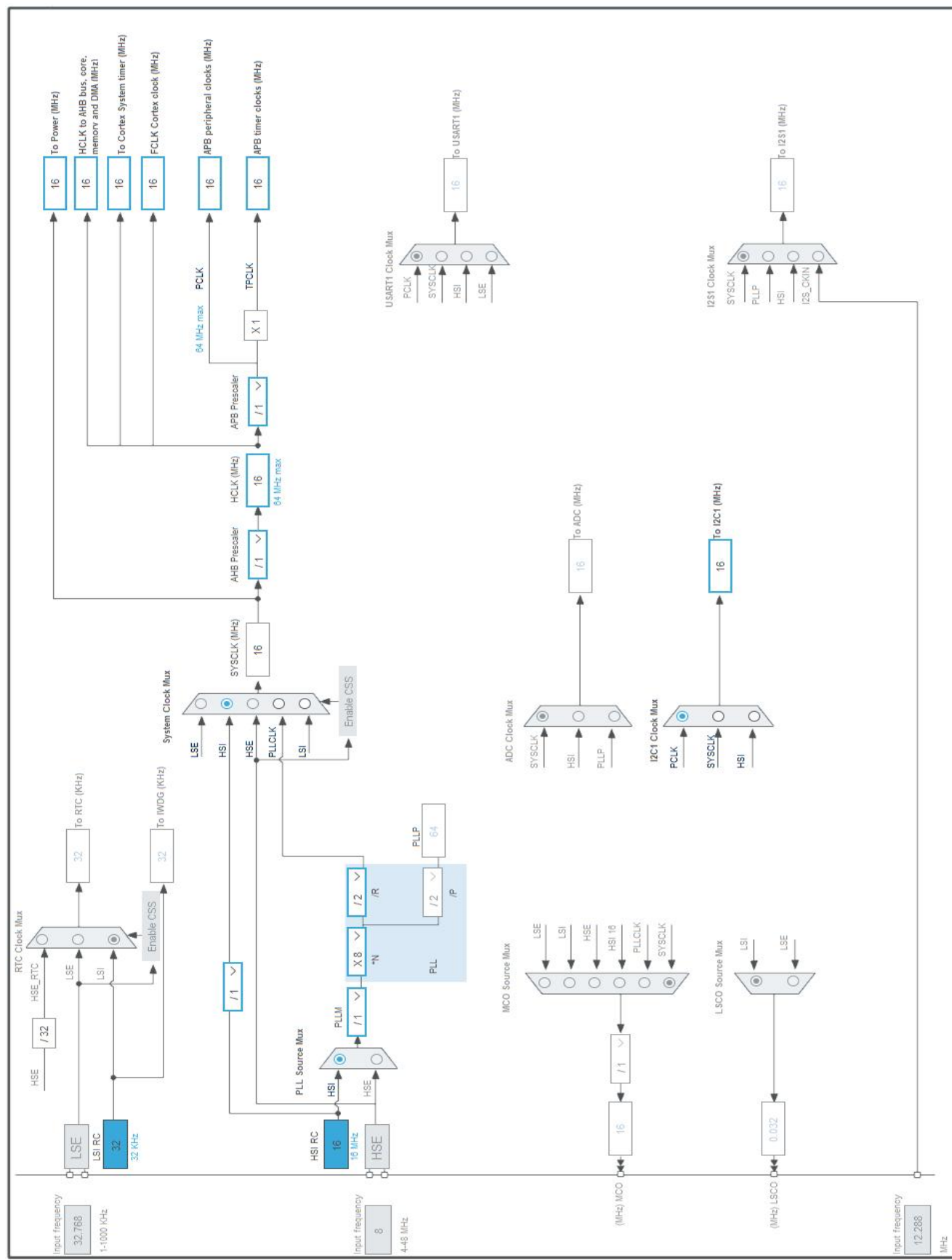
## 2. Pinout Configuration



### 3. Pins Configuration

| Pin Number<br>TSSOP20 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label |
|-----------------------|---------------------------------------|----------|--------------------------|-------|
| 1                     | PB8                                   | I/O      | I2C1_SCL                 |       |
| 2                     | PB9                                   | I/O      | I2C1_SDA                 |       |
| 4                     | VDD/VDDA                              | Power    |                          |       |
| 5                     | VSS/VSSA                              | Power    |                          |       |
| 6                     | NRST                                  | Reset    |                          |       |
| 18                    | PA13                                  | I/O      | SYS_SWDIO                |       |
| 19                    | PA14-BOOT0                            | I/O      | SYS_SWCLK                |       |

## 4. Clock Tree Configuration



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |                      |
|-----------|----------------------|
| Series    | STM32G0              |
| Line      | STM32G0x0 Value line |
| MCU       | STM32G030F6Px        |
| Datasheet | DS12991 Rev1         |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.0 |

### 1.3. Battery Selection

|                   |                  |
|-------------------|------------------|
| Battery           | Li-SOCL2(AAA700) |
| Capacity          | 700.0 mAh        |
| Self Discharge    | 0.08 %/month     |
| Nominal Voltage   | 3.6 V            |
| Max Cont Current  | 10.0 mA          |
| Max Pulse Current | 30.0 mA          |
| Cells in series   | 1                |
| Cells in parallel | 1                |

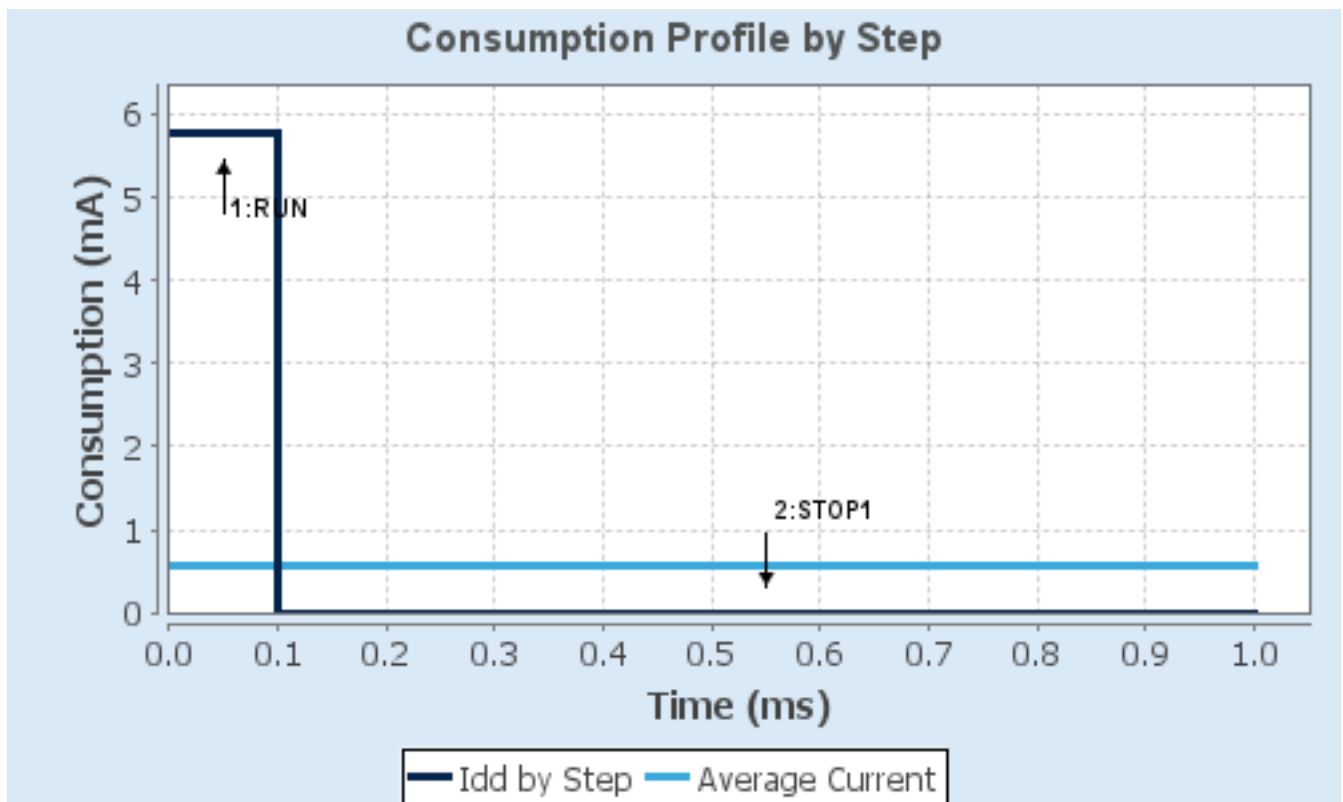
#### 1.4. Sequence

|                               |             |                 |
|-------------------------------|-------------|-----------------|
| <b>Step</b>                   | Step1       | Step2           |
| <b>Mode</b>                   | RUN         | STOP1           |
| <b>Vdd</b>                    | 3.0         | 3.0             |
| <b>Voltage Source</b>         | Battery     | Battery         |
| <b>Range</b>                  | Range1-High | NoRange         |
| <b>Fetch Type</b>             | FLASH       | Flash-PowerDown |
| <b>CPU Frequency</b>          | 64 MHz      | 16 MHz          |
| <b>Clock Configuration</b>    | HSI PLL     | HSI             |
| <b>Clock Source Frequency</b> | 16 MHz      | 16 MHz          |
| <b>Peripherals</b>            |             |                 |
| <b>Additional Cons.</b>       | 0 mA        | 0 mA            |
| <b>Average Current</b>        | 5.77 mA     | 3.7 $\mu$ A     |
| <b>Duration</b>               | 0.1 ms      | 0.9 ms          |
| <b>DMIPS</b>                  | 80.0        | 0.0             |
| <b>Ta Max</b>                 | 128.68      | 130             |
| <b>Category</b>               | In DS Table | In DS Table     |

#### 1.5. Results

|               |                               |                 |                |
|---------------|-------------------------------|-----------------|----------------|
| Sequence Time | 1 ms                          | Average Current | 580.33 $\mu$ A |
| Battery Life  | 1 month, 19 days,<br>18 hours | Average DMIPS   | 80.0 DMIPS     |

#### 1.6. Chart



## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                  |
|-----------------------------------|------------------------|
| Project Name                      | AHT30_lib              |
| Project Folder                    | D:\STM32Proj\AHT30_lib |
| Toolchain / IDE                   | STM32CubeIDE           |
| Firmware Package Name and Version | STM32Cube FW_G0 V1.6.2 |
| Application Structure             | Advanced               |
| Generate Under Root               | Yes                    |
| Do not generate the main()        | No                     |
| Minimum Heap Size                 | 0x200                  |
| Minimum Stack Size                | 0x400                  |

### 2.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                    |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name      | Peripheral Instance Name |
|------|--------------------|--------------------------|
| 1    | SystemClock_Config | RCC                      |
| 2    | MX_GPIO_Init       | GPIO                     |
| 3    | MX_I2C1_Init       | I2C1                     |

## 3. Peripherals and Middlewares Configuration

### 3.1. I2C1

#### I2C: I2C

##### 3.1.1. Parameter Settings:

###### Timing configuration:

|                               |                     |
|-------------------------------|---------------------|
| Custom Timing                 | Disabled            |
| I2C Speed Mode                | Standard Mode       |
| I2C Speed Frequency (KHz)     | 100                 |
| Rise Time (ns)                | 100                 |
| Fall Time (ns)                | 100                 |
| Coefficient of Digital Filter | 0                   |
| Analog Filter                 | Enabled             |
| Timing                        | <b>0x00503D58 *</b> |

###### Slave Features:

|                                  |          |
|----------------------------------|----------|
| Clock No Stretch Mode            | Disabled |
| General Call Address Detection   | Disabled |
| Primary Address Length selection | 7-bit    |
| Dual Address Acknowledged        | Disabled |
| Primary slave address            | 0        |

### 3.2. RCC

##### 3.2.1. Parameter Settings:

###### System Parameters:

|                   |                    |
|-------------------|--------------------|
| VDD voltage (V)   | 3.3                |
| Instruction Cache | Enabled            |
| Prefetch Buffer   | Enabled            |
| Data Cache        | Enabled            |
| Flash Latency(WS) | 0 WS (1 CPU cycle) |

###### RCC Parameters:

|                                |      |
|--------------------------------|------|
| HSI Calibration Value          | 64   |
| HSE Startup Timeout Value (ms) | 100  |
| LSE Startup Timeout Value (ms) | 5000 |

###### Power Parameters:

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 1 |
|-------------------------------|---------------------------------|

###### Peripherals Clock Configuration:

|                                              |      |
|----------------------------------------------|------|
| Generate the peripherals clock configuration | TRUE |
|----------------------------------------------|------|

### 3.3. SYS

mode: Debug

Timebase Source: SysTick

\* User modified value

## 4. System Configuration

### 4.1. GPIO configuration

| IP   | Pin        | Signal    | GPIO mode                     | GPIO pull/up pull down      | Max Speed | User Label |
|------|------------|-----------|-------------------------------|-----------------------------|-----------|------------|
| I2C1 | PB8        | I2C1_SCL  | Alternate Function Open Drain | No pull-up and no pull-down | Low       |            |
|      | PB9        | I2C1_SDA  | Alternate Function Open Drain | No pull-up and no pull-down | Low       |            |
| SYS  | PA13       | SYS_SWDIO | n/a                           | n/a                         | n/a       |            |
|      | PA14-BOOT0 | SYS_SWCLK | n/a                           | n/a                         | n/a       |            |

### 4.2. DMA configuration

nothing configured in DMA service

### 4.3. NVIC configuration

#### 4.3.1. NVIC

| Interrupt Table                                                           | Enable | Preenmption Priority | SubPriority |
|---------------------------------------------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                                                    | true   | 0                    | 0           |
| Hard fault interrupt                                                      | true   | 0                    | 0           |
| System service call via SWI instruction                                   | true   | 0                    | 0           |
| Pendable request for system service                                       | true   | 0                    | 0           |
| System tick timer                                                         | true   | 3                    | 0           |
| Flash global interrupt                                                    | unused |                      |             |
| RCC global interrupt                                                      | unused |                      |             |
| I2C1 event global interrupt / I2C1 wake-up interrupt through EXTI line 23 | unused |                      |             |

#### 4.3.2. NVIC Code generation

| Enabled interrupt Table                 | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-----------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                  | false                             | true                 | false            |
| Hard fault interrupt                    | false                             | true                 | false            |
| System service call via SWI instruction | false                             | true                 | false            |
| Pendable request for system service     | false                             | true                 | false            |
| System tick timer                       | false                             | true                 | true             |

\* User modified value

## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

#### Middleware

#### System Core

#### Analog

#### Timers

#### Connectivity

#### Multimedia

#### Computing

DMA

I2C1



GPIO



IIVIC



RCC



SYS



## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                        |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32g0_ibis.zip">https://www.st.com/resource/en/ibis_model/stm32g0_ibis.zip</a>                                                                                                                                                         |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32g0-svd.zip">https://www.st.com/resource/en/svd/stm32g0-svd.zip</a>                                                                                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                 |
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| Brochures               | <a href="https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf">https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf</a>                                                                                       |
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| Flyers                  | <a href="https://www.st.com/resource/en/flyer/fldpstpfc11120.pdf">https://www.st.com/resource/en/flyer/fldpstpfc11120.pdf</a>                                                                                                                                                               |
| Security Bulletin       | <a href="https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-">https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-</a> |

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